

General Mathematics Reviewer

Topic: Functions and Relations

Discussion

- Relation: A set of ordered pairs (x, y) .
- Function: A special relation where each x (input) is paired with only one y (output).
- Not a function: If a single x corresponds to more than one y .
- Vertical Line Test: If a vertical line touches the graph more than once, the relation is not a function.
- Inverse of a function: Obtained by swapping x and y , then solving for y . Not all functions have inverses that are functions (e.g., $y=x^2$ is not one-to-one).
- Function operations: You can add, subtract, multiply, divide, and compose functions.
 - Composition: $(f \circ g)(x) = f(g(x))$.
- Zeros of a function: Values of x where $f(x)=0$.

PRACTICE QUESTIONS

1. What makes a relation a function?
- a. Each input has exactly one output
 - b. The range values must be positive
 - c. Each output has two inputs
 - d. The graph is symmetric to the x -axis
 - e. The domain values must be integers

ANSWER: **a. Each input has exactly one output** → This is the definition of a function.

2. Which of the following passes the vertical line test?
- a. Circle at origin radius 2 → **Fails vertical line test**
 - b. Sideways parabola → **Fails vertical line test**
 - c. Upward parabola
 - d. Ellipse → **Fails vertical line test**
 - e. Vertical line → **Fails vertical line test**

ANSWER: **c. Upward parabola** → Passes vertical line test.

3. If $f(x)=x^2$, $g(x)=x+1$, find $(f \circ g)(2)$.

Solution:

4. If $f(x)=3x-2$, $g(x)=x+4$, find $(f+g)(x)$.

Solution:

5. If $f(x)=x+3$, $g(x)=x-5$, find $(f \bullet g)(x)$.

Solution:

6. If $f(x)=x^2$, $g(x)=x+3$, evaluate $(f-g)(2)$.

Solution:

7. If $f(x)=2x$, $g(x)=x^2$, find $(g\circ f)(2)$.

Solution:

8. If $f(x)=x^2$, $g(x)=x-1$, find $(f\circ g)(x)$.

Solution:

9. Find the inverse of $f(x)=2x+7$.

Solution:

10. If $f(x)=\frac{x-3}{2}$, find $f^{-1}(5)$.

Solution

11. Which function has no inverse that is also a function?

- a. $f(x)=x^2$
- b. $f(x)=3x-4$
- c. $f(x)=x^3$
- d. $f(x)=\sqrt{x}$
- e. $f(x)=\frac{1}{x}$

12. If $f(x)=\frac{x+2}{x-1}$, identify its zero and vertical asymptote.

Solution:

13. What does $(f \circ g)(x)$ mean?
- Add $f(x)$ and $g(x)$
 - Multiply $f(x)$ and $g(x)$
 - Substitute $g(x)$ into $f(x)$
 - Substitute x into both functions
 - Substitute $f(x)$ into $g(x)$
14. Which of the following sets represents a function?
- $\{(1,2), (1,3)\}$
 - $\{(2,3), (3,4), (4,5)\}$
 - $\{(0,1), (0,2)\}$
 - $\{(5,1), (6,2), (5,3)\}$
 - $\{(7,1), (7,2), (7,3)\}$
15. After switching x and y in $y=3^x$, what's the next step to find its inverse?
- Multiply both sides by 3
 - Take log base 3 of both sides
 - Add 3 to both sides
 - Divide both sides by 3
 - Replace y with $f^{-1}(x)$
16. What does it mean when a number is a zero of a function?
- It is where the graph cuts the x -axis
 - It is where the graph cuts the y -axis
 - It is the minimum of the graph
 - It is the slope of the graph
 - It is the range value of the function
17. How do you find the zeros of $f(x)=\frac{p(x)}{q(x)}$?
- Solve denominator=0
 - Solve numerator=0
 - Solve both numerator and denominator=0
 - Solve where function is undefined
 - Multiply numerator and denominator

Answers with Explanations

- a – Definition of function.
- c – Only parabola passes vertical line test.
- 9 – $f(g(2))=f(3)=9$.
- $4x+2$ – Simplified sum of functions.
- $x^2-2x-15$ – Multiplying binomials.
- $-1 - 4 - 5 = -10$.
- $8 - 4 + 4 = 8$.
- x^2+x-1 – Simplified sum.
- $f^{-1}(x)=\frac{x-7}{2}$.
- 13 – Substitute and solve.
- a – x^2 is not one-to-one.
- Zero: -2, Vertical asymptote: 1.

13. c – Composition definition.

14. b – Each input has exactly one output.

15. b – Apply log base 3.

16. a – Zero is x-intercept.

17. b – Zeros come from numerator.